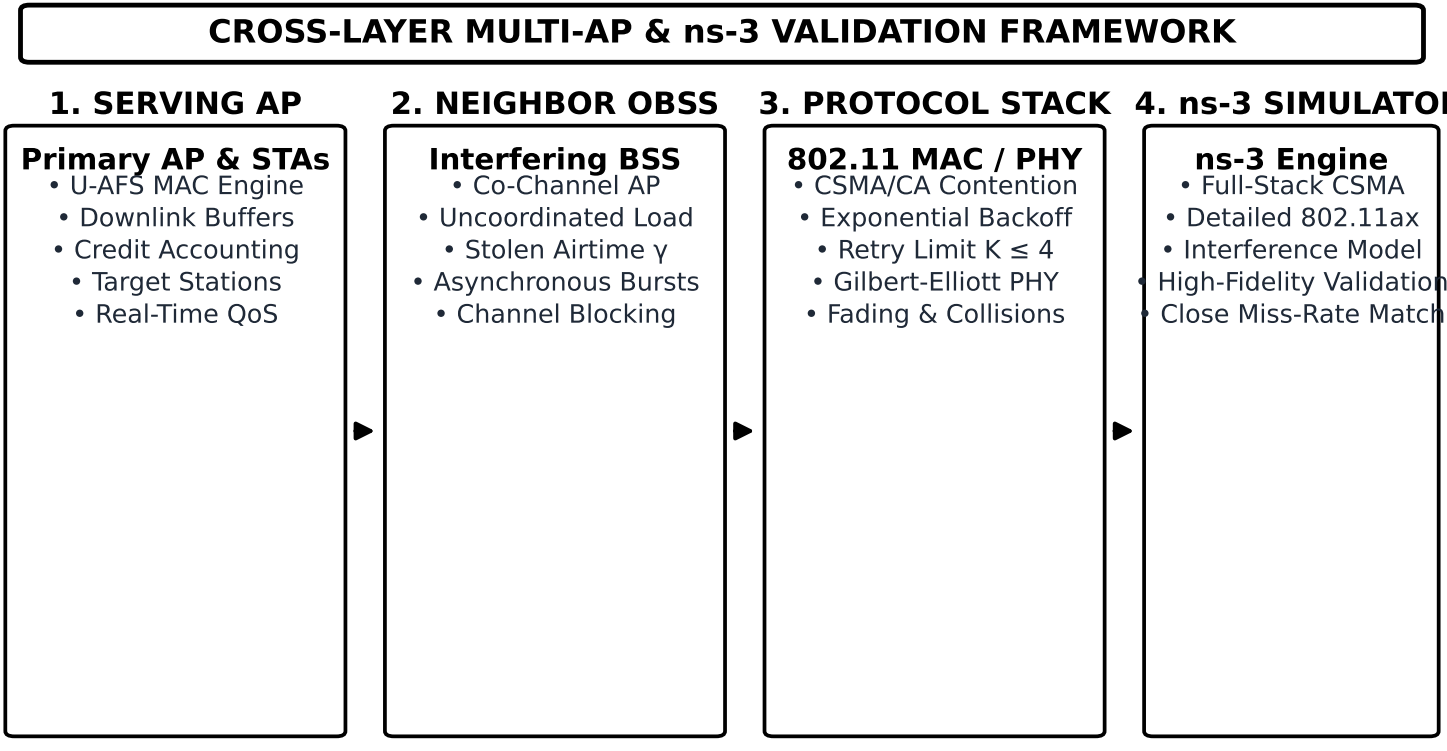
CROSS-LAYER MULTI-AP & ns-3 VALIDATION FRAMEWORK
1. SERVING AP
Primary AP & STAs
• U-AFS MAC Engine
• Downlink Buffers
• Credit Accounting
• Target Stations
• Real-Time QoS
2. NEIGHBOR OBSS
Interfering BSS
• Co-Channel AP
• Uncoordinated Load
• Stolen Airtime γ
• Asynchronous Bursts
• Channel Blocking
3. PROTOCOL STACK
802.11 MAC / PHY
• CSMA/CA Contention
• Exponential Backoff
• Retry Limit K ≤ 4
• Gilbert-Elliott PHY
• Fading & Collisions
4. ns-3 SIMULATOR
ns-3 Engine
• Full-Stack CSMA
• Detailed 802.11ax
• Interference Model
• High-Fidelity Validation
• Close Miss-Rate Match
Cross-Validation: Confirms U-AFS Superiority Across Realistic Channel Contention